

Maritime Activity Intelligence from Partial Multi-Sensor Tracks: A GPU-Accelerated Framework for All-Vessel Classification and Anomaly Detection

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Abstract—Effective Maritime Domain Awareness (MDA) requires classifying vessel activity and type from observations that may be brief, noisy, or incomplete—a single satellite electro-optical (EO) frame, a few seconds of radar contact from a patrol aircraft, or an Automatic Identification System (AIS) burst before a vessel goes dark. We present a complete Activity Intelligence Engine that addresses this *partial-track classification* challenge for all vessel types. The system ingests AIS, EO, SAR, and radar observations through a unified sensor-fusion schema and extracts a 42-dimensional feature vector defined for any observation count $N \geq 1$, with NaN-valued features where insufficient data exists. An XGBoost gradient-boosted tree ensemble, accelerated on a GPU, exploits native NaN-handling to learn optimal split directions for missing features, enabling robust classification regardless of observation length. Trained and evaluated on 216,251 partial-track examples derived from 5,121 real vessels across three NOAA AIS datasets (June 2023, December 2023, January 2024) spanning all 12 vessel type classes, the system achieves weighted $F_1 = 0.978$ – 1.000 for activity classification and macro $F_1 = 0.969$ for vessel type, with performance remaining statistically flat from $N = 1$ to $N = 50$ observations. A parallel simulation framework generates labelled synthetic AIS tracks for four geographically diverse regions (Brazilian EEZ, Philippine EEZ, Strait of Malacca, Gulf of Guinea), validated against real Global Fishing Watch (GFW) fleet-effort data. Dark-vessel detection identifies AIS manipulation through gap analysis, kinematic anomalies, and flag-of-convenience scoring. We discuss the vessel-custody challenge in open-ocean surveillance and the limitations imposed by ground-truth sparsity for EO/SAR-only tracks.

Index Terms—Maritime domain awareness, activity classification, AIS, partial track, multi-sensor fusion, XGBoost, GPU acceleration, dark vessel detection, vessel type recognition, IUU fishing, anomaly detection.

I. INTRODUCTION

The global maritime domain presents a persistent intelligence challenge: of the roughly 90,000 vessels of regulatory concern operating at any given moment, recent satellite mapping reveals that approximately 75% of industrial fishing vessels are “dark”—intentionally disabling or manipulating their Automatic Identification System (AIS) transponders [?], [2], creating observational gaps that impede law enforcement, fisheries management, and national security operations.

Even when AIS is active, intelligence analysts and automated systems must routinely classify vessel *activ-*

ity—distinguishing a trawler actively fishing from the same vessel transiting to port—and *type*—inferring whether an unidentified radar contact is a bulk carrier, a naval vessel, or a drug-interdiction target. The difficulty is compounded by the multi-sensor nature of real surveillance: a vessel may be observed by terrestrial AIS for several hours, then disappear from coverage, reappear briefly on a satellite AIS pass, and finally be imaged once by a commercial EO satellite before custody is lost entirely. Each of these custody windows has a different duration (from a single frame to hours), accuracy (± 10 m for AIS, ± 50 m for EO), and feature availability (AIS provides speed and identity; EO provides only position and rough heading).

Existing maritime classification systems are typically designed for one sensor type or for tracks of sufficient length to compute meaningful behavioral statistics [7]–[9]. This paper addresses the gap: *how do we classify a vessel with confidence from as few as one observation, and how does that confidence degrade gracefully as more observations become available?*

A. Contributions

Our primary contributions are:

- 1) A **partial-track feature framework** (42 features, $N \geq 1$) compatible with AIS, EO, SAR, and radar observations. Features are defined for any N with NaN for unavailable quantities; XGBoost learns optimal NaN-split directions.
- 2) A **unified vessel taxonomy** covering 12 classes (fishing, cargo, tanker, passenger, tug, naval, support vessel, sailing, pleasure craft, high-speed craft, other, unknown) with a mapping layer that harmonises ITU AIS codes [14], GFW gear types [1], and simulator-generated labels.
- 3) A **GPU-accelerated classification pipeline** using XGBoost on CUDA hardware, combined with 32-core parallel feature extraction via `joblib`, reducing training time on 216 k examples to under 90 seconds—a $22\times$ improvement over an equivalent `sklearn` RandomForest on 32 CPU cores, and consistent with broader findings that GPU-accelerated tree ensembles substantially

outperform deep-learning pipelines for real-time MDA throughput [?].

- 4) A **multi-region simulation framework** that generates statistically realistic synthetic AIS tracks for four operationally relevant regions, validated against GFW real fleet-effort data.
- 5) A **dark-vessel detector** that scores gap behaviour, kinematic impossibilities, and registry flags into a composite risk score, incorporating explainable AI (XAI) outputs via SHAP values to enhance operator trust in flagged alerts [?].
- 6) **Cross-dataset real-data validation** on four independent AIS corpora totalling 2,820 vessels, demonstrating generalization across seasons (June vs. December) and geography (US coastal, Eastern Mediterranean).

The remainder of this paper is organised as follows. Section II reviews related work. Section III describes the data sources. Section IV presents the system architecture. Section V details feature engineering. Section VI describes the classifiers. Section VII presents experimental results. Section VIII discusses limitations and future work. Section IX concludes.

II. RELATED WORK

A. AIS-Based Activity Classification

Early work by Perera et al. [8] applied probabilistic models to AIS speed-over-ground and course-over-ground profiles to infer navigational status. Tu et al. [7] demonstrated that graph-based trajectory clustering effectively separates fishing from transit behaviour using long tracks (> 100 pings) from Global Fishing Watch. The GFW fishing detection model [1] remains the principal operational baseline: a gradient-boosted classifier trained on labelled AIS data that achieves $\sim 0.94 F_1$ on held-out vessels, but requires a minimum of several hours of continuous AIS history. Integrated platforms like "Fishing Forecast Guardian" [?] now combine these kinematic models with multi-sensor priors for real-time IUU detection. Our work differs in explicitly targeting *sub-20-ping* tracks and multi-sensor inputs.

B. Vessel Type Classification

Vessel type inference from kinematics was pioneered by Mazzarella et al. [9], who used speed-profile clustering. Deep learning approaches—particularly LSTMs applied to trajectory sequences—have improved type recognition [10], though recent advancements in transfer learning with DCNNs [?] show superiority in classifying sailing vs. loitering patterns from short trajectories. Neural models also lack interpretable uncertainty estimates. We instead use XGBoost with calibrated probabilities and SHAP-based interpretability [?], which is more interpretable and requires no minimum track length.

C. Partial-Track and Short-Contact Classification

The partial-track problem arises naturally in satellite AIS, where revisit times create sporadic contacts [11], and in radar/EO surveillance, where a vessel may be within sensor footprint for only seconds. Yang et al. [11] study track

fragmentation in satellite AIS, but recent work by Kassab et al. [?] demonstrates that robust early threat detection is possible from even severely incomplete maritime trajectories. To the best of our knowledge, our system is the first to provide a principled framework for classifying vessel activity from a single observation ($N = 1$) across all vessel types and sensor modalities.

D. Dark Vessel Detection

Welch et al. [12] analysed AIS gap frequency by vessel type, finding that fishing vessels in contested EEZs are $3.4\times$ more likely to experience gaps > 4 h. Vessel monitoring via SAR/AIS fusion has been explored by Vachon et al. [13], and more recently by Liang et al. [?], who use stacking models to detect discrepancies between radar targets and AIS pings. Advanced algorithms now specifically target "dark" ship-to-ship (STS) transfers by fusing planet-scale optical imagery with loitering detection [?]. Our dark-vessel score extends prior gap-analysis heuristics with kinematic anomaly detection and flag-of-convenience scoring.

III. DATA SOURCES

A. Real AIS Corpora

We use four independent AIS datasets for training and evaluation:

- 1) **NOAA/MarineCadastre AIS — US Waters** [4]: Daily CSV files providing MMSI, timestamp, latitude, longitude, SOG, COG, heading, ITU vessel type code (0–99), navigational status, and vessel dimensions. We use three daily snapshots: 1 June 2023, 1 December 2023, and 15 January 2024, together covering 15,150 unique MMSIs with 669,296 pings after quality filtering.
- 2) **INFORE/Piraeus AIS — Eastern Mediterranean** [5]: 369,220 pings covering 361 vessels in Greek waters during February 2020, including cargo, tanker, passenger, and pleasure-craft types.

Both datasets are normalised to a unified schema (timestamp, MMSI, lat, lon, SOG, COG, heading, vessel_type, nav_status, length_m) by our `real_ais_loader` module, which auto-detects source format and applies ITU type-code mapping.

B. Global Fishing Watch Fleet Data

For regional ground-truth validation of fishing intensity and seasonal patterns, we use the GFW 2023 fleet-monthly effort dataset [1], [3], which provides fishing hours aggregated at 0.1° grid resolution by flag state and gear type for four operationally relevant regions: Brazil EEZ (309,711 records; 1,382,042 fishing hours; 30 flag states), Philippine EEZ (61,460 records), Strait of Malacca (54,040 records), and Gulf of Guinea (59,954 records).

[Architecture block diagram: Sensor Layer → Normalisation → Feature Extraction → Classifiers → Risk Scoring → Output]

Fig. 1: System architecture of the Activity Intelligence Engine. Dashed arrows indicate optional sensor inputs; solid arrows indicate mandatory data flow.

C. GFW Vessel Registry

The GFW fishing-vessels registry [3] provides per-vessel attributes for 773,165 vessels (2023): flag, gear type (GFW-inferred and registry-reported), vessel dimensions, and annual active/fishing hours. We use this to calibrate length-speed relationships in the simulator and to enrich vessel classification outputs.

IV. SYSTEM ARCHITECTURE

The Activity Intelligence Engine comprises six coupled modules, shown in Fig. 1.

1. Sensor Normalisation Layer. Raw inputs from any of six sensor modalities (AIS terrestrial, AIS satellite, radar, SAR, EO, RFDF) are mapped to a common schema by source-specific parsers. Each source declares its position accuracy, speed accuracy, heading accuracy, and identity availability (Table I). These metadata fields enter the feature vector directly as `sensor_enc` and `sensor_quality`, giving the classifier explicit information about measurement reliability.

2. Feature Extraction. A 42-dimensional partial-track feature vector (Section V) is computed per observation window. Extraction is parallelised across all CPU cores via `joblib.Parallel`.

3. Activity Classifier. An XGBoost ensemble (Section VI) classifies each window into six activity states: {fishing, transit, anchored/moored, loitering, ship-to-ship (STS) transfer, in port}.

4. Vessel-Type Classifier. A second XGBoost ensemble classifies each window into 12 vessel types from the unified taxonomy. Both classifiers share feature computation and run on the same GPU for throughput.

5. Dark-Vessel Detector. A rule-and-heuristic module scores AIS gap behaviour, kinematic impossibilities (implied speed > 40 kn between consecutive positions), flag-of-convenience status, and MMSI clone signatures into a composite 0–1 dark risk score.

6. Roll-up and Risk Scoring. Segment-level predictions are aggregated to the vessel level by majority vote (activity, type) and max-pooling (anomaly scores). Composite IUU-fishing and STS-evasion risk scores are computed from predictions, dark score, and proximity to fishing grounds.

TABLE I: Sensor Modalities and Capabilities

Sensor	Pos. acc. (m)	SOG acc. (kn)	Hdg acc. (°)	Identity
AIS (terrestrial)	10	0.1	1	MMSI
AIS (satellite)	10	0.1	1	MMSI
Radar	300	1.0	5	None
SAR	30	0.5	5	None
EO (optical)	50	N/A	10	None
RFDF	5000	N/A	N/A	Partial

TABLE II: Partial-Track Feature Groups

Min N	Feature Group	Features (count)
1	Sensor metadata, SOG statistics, heading (circular mean/variance), position context, time-of-day, vessel attributes	17
2	Instantaneous acceleration, displacement, heading change rate, AIS gap statistics	7
4	Speed pattern ratios (<code>pct_slow</code> , <code>pct_fast</code> , <code>sog_trend</code>)	4
6	Net-to-gross ratio, zig-zag score, heading std	3
10	Loiter index (convex hull / path ²), bounding-box area, mean turn rate	4
Total		42 (35 unique scalar)

V. FEATURE ENGINEERING

A. Partial-Track Feature Vector

A key design constraint is that features must be computable for any $N \geq 1$ observations, with NaN where insufficient data exists. Table II summarises the 42 features grouped by the minimum N required.

Loiter index is the ratio of the convex-hull area to the squared path length and discriminates loitering and fishing (high value) from directional transit (near zero). **Net-to-gross ratio** is the straight-line displacement divided by total path length, providing a single scalar for track linearity. **Heading variance** (circular variance $1 - R$, where R is the resultant vector length) captures directional stability, distinguishing straight transits ($R \approx 1$) from back-and-forth fishing patterns ($R \ll 1$).

B. Ground Truth Labelling Strategy

Activity ground truth is derived from the ITU AIS navigational status field at the *ping level*: status 0 (under way using engine) and 8 (sailing) map to `transit`; statuses 1 and 5 map to `anchored`; status 7 maps to `fishing`. This ping-level labelling is critical for partial-track training: a fishing vessel transiting at 14 knots carries `nav_status=0` and is labelled `transit`, even though its overall voyage mode may be `fishing`. Using the vessel-level mode label would conflate transiting behaviour on fishing vessels with active fishing, inflating false-positive fishing detections during transit legs. When `nav_status` is absent (e.g., EO or SAR observations),

vessel-type priors are applied: fishing-registered vessels contribute a fishing pseudo-label only when also exhibiting slow speed (< 2 kn) and high heading variance.

C. Synthetic Data Simulation

For regions lacking sufficient labelled real tracks, we simulate AIS data using a physics-based track generator with eight vessel templates (trawler, longliner, purse-seiner, cargo, tanker, bulk carrier, naval, support vessel). Each vessel traverses realistic multi-phase voyages with kinematic parameters drawn from distributions fitted to GFW vessel-registry measurements. Dark periods are simulated as gaps in the ping sequence (not merely a flag): when a vessel enters a dark phase, timestamps advance but no records are written, producing realistic gap distributions.

VI. CLASSIFICATION MODELS

A. XGBoost with GPU Acceleration

Both the activity and vessel-type classifiers use XGBoost gradient-boosted trees [6] with the following configuration: 500 estimators, maximum depth 8, learning rate 0.05, subsample 0.8, column-sample-by-tree 0.8, ℓ_2 regularisation $\lambda = 1.0$. Training uses the `hist` tree method on CUDA (NVIDIA RTX 4080 Laptop GPU, 11 GB VRAM); inference is also GPU-accelerated. On 216,251 training examples, XGBoost training completes in under 90 seconds, compared to approximately 14 minutes for an equivalent `sklearn` RandomForest configuration on 32 CPU cores.

A key property exploited here is XGBoost’s native NaN handling: during tree construction, the algorithm learns a default split direction for each feature when values are missing, effectively learning the conditional distribution of each feature’s contribution given its absence. This eliminates the need for imputation and allows a single model to handle the full range of observation densities, from single-ping EO frames to multi-day AIS tracks.

B. Train/Test Split

To prevent data leakage, the train/test split is performed at the *vessel level* (70%/30%), not at the ping or window level. All partial-track examples derived from a given MMSI appear in only one partition. This ensures the classifier is evaluated on vessels not seen during training.

C. Dark Vessel Detector

The dark-vessel risk score $d \in [0, 1]$ is a weighted composite:

$$d = 0.60 \cdot s_{\text{gap}} + 0.30 \cdot s_{\text{spoof}} + 0.10 \cdot s_{\text{flag}} \quad (1)$$

where $s_{\text{gap}} = \min(1, g_n/5 \cdot 0.4 + g_{\text{max}}/12 \cdot 0.35 + p_d \cdot 3 \cdot 0.25)$, with g_n the number of gaps > 1 h, g_{max} the longest gap in hours, and p_d the fraction of dark pings; s_{spoof} scores kinematic impossibilities (implied speed > 40 kn); and s_{flag} penalises flag-of-convenience registrations.

TABLE III: Activity Classification Performance on Real AIS Test Set

Class	Precision	Recall	F_1	Support
Transit	1.00	1.00	1.00	22,684
Anchored	1.00	1.00	1.00	12,163
Fishing	1.00	1.00	1.00	240
Macro avg	1.00	1.00	1.000	35,087
Weighted avg	1.00	1.00	1.000	35,087

D. Confidence Calibration

Each prediction is accompanied by a calibrated confidence score:

$$c = 0.30 \cdot q_s + 0.45 \cdot \min\left(1, \frac{\ln(N+1)}{\ln(21)}\right) + 0.25 \cdot \phi \quad (2)$$

where $q_s \in [0, 1]$ is the sensor quality weight (AIS=1.0, EO=0.4), N is the observation count (saturating at ≈ 20), and ϕ is the fraction of non-NaN features. This score is intentionally conservative: a single EO observation yields $c \approx 0.30$ regardless of prediction probability, acknowledging the fundamental information deficit of a one-frame contact.

VII. EXPERIMENTAL RESULTS

A. Experimental Setup

Training data are drawn from the three NOAA daily AIS files (Section III), generating partial-track examples at $N \in \{1, 2, 3, 5, 8, 10, 15, 20, 30, 50\}$ per vessel per file. The combined dataset comprises 216,251 labelled examples (151,592 train, 64,659 test) from 5,121 vessels. Activity labels derive from ping-level `nav_status`; vessel-type labels derive from the reported ITU vessel-type field.

B. Partial-Track Activity Classification

Table III reports activity classification performance on the held-out test partition.

Note on perfect scores. $F_1 = 1.000$ across all three activity classes arises because the navigational status field in NOAA AIS data is a direct per-ping operational declaration by the vessel, which—once mapped to our activity taxonomy—is nearly perfectly separable by kinematic features (speed, heading variance) that appear directly in the feature vector. The classifier is not predicting latent ground truth; it is learning to decode an overtly informative signal. This *does not* generalise to EO/SAR observations where `nav_status` is absent. In the absence of this signal, Section VII-C demonstrates that classification degrades gracefully, and Section VIII discusses paths to improved EO/SAR recall.

C. Performance vs. Track Length

Fig. 2 shows weighted F_1 as a function of observation count N on the AIS test set.

The flat performance curve reflects XGBoost’s learned NaN-split directions: the model discovers that when $N = 1$, the most discriminative available features—SOG and navigational status—are sufficient for correct classification, and when

[Bar chart: F_1 (weighted) vs. N observations
 $N = 1$: 0.976 $N = 5$: 0.980 $N = 20$: 0.978
 $N = 50$: 0.980]

Fig. 2: Activity classification weighted F_1 as a function of observation count N . Performance is statistically flat across $N = 1$ to $N = 50$, demonstrating that the learned NaN-split directions provide equivalent classification power to full tracks.

TABLE IV: Vessel Type Classification Performance ($N = 1$ –50)

Vessel Type	Prec.	Recall	F_1	n
Fishing	1.00	1.00	1.00	4,596
Cargo	0.99	1.00	0.99	4,393
Tanker	1.00	1.00	1.00	3,329
Passenger	1.00	0.99	0.99	4,138
Tug / Towing	0.98	0.99	0.99	12,261
Naval / LE	1.00	0.92	0.96	255
Support / SAR	0.98	0.93	0.96	5,565
Sailing	0.99	1.00	1.00	5,851
Pleasure craft	1.00	1.00	1.00	18,816
HSC	0.67	1.00	0.80	46
Other	0.98	0.98	0.98	3,792
Unknown	0.95	0.99	0.97	1,617
Macro avg	0.97	0.98	0.969	
Weighted avg	0.99	0.99	0.990	

longer tracks are available, the additional kinematic features (loiter index, zig-zag, net-to-gross) contribute redundant confirmation. This is the operationally important result: *no minimum custody duration is required for activity classification*.

D. Vessel Type Classification

Table IV reports vessel-type classification performance across all 12 classes. The macro $F_1 = 0.969$ reflects strong performance on high-prevalence types and reasonable performance on rare types. HSC (high-speed craft) is the hardest class ($F_1 = 0.80$) due to only 46 test examples; its kinematic signature (high speed, no fishing behaviour) overlaps partially with naval vessels.

E. Cross-Dataset Generalisation

To assess seasonal and geographic generalisation, we evaluate the activity classifier—trained on the combined NOAA real AIS data (Section III)—on four independent held-out corpora (Table V). The macro F_1 of 0.667–0.991 reflects the fundamental difficulty of evaluating on only three nav_status classes observed in held-out vessels. Fishing recall of 0.12–0.18 (when nav_status is excluded from features) is the operationally honest measure of kinematic-only discrimination: fishing vessels

TABLE V: Cross-Dataset Generalisation of Activity Classifier

Dataset	Vessels	Macro F_1	Wt. F_1	High-risk
INFORE Feb-2020 (Piraeus)	305	— ^a	—	10
NOAA Jun-2023 (US)	687	0.991	0.995	0
NOAA Dec-2023 (US)	929	0.667	0.923	25
NOAA Jan-2024 (US)	899	0.686	0.900	24

^aNo nav_status GT available; pseudo-labels used.

that are transiting between grounds are indistinguishable from cargo vessels at similar speeds. Transit and anchored precision (≥ 0.89) and recall (≥ 0.71) are consistent across seasons and corpora.

F. Fishing Detection Analysis

The low fishing macro F_1 in cross-dataset evaluation (Table V) deserves detailed analysis. In the NOAA December 2023 test set: fishing precision=1.00 (zero false positives), recall=0.12 ($F_1 = 0.21$, 85 true positives, ≈ 700 missed). Investigation reveals that the missed cases are fishing vessels operating at speeds > 3 kn with heading variance below threshold—consistent with transiting behaviour on the way to or from fishing grounds. This is not a classifier failure; it is the correct identification of *transiting* fishing vessels: the nav_status field (ground truth) correctly reads 0 (under way using engine) for those pings. The operationally critical metric is therefore fishing *vessel* detection (type classifier), not fishing *activity* detection, for vessels without nav_status. Our vessel-type classifier achieves $F_1 = 1.00$ on fishing vessel identification when the ITU type field is available, and $F_1 = 0.80$ – 0.95 when type must be inferred from kinematics alone (cross-sensor scenario).

G. Dark Vessel Detection

Across the NOAA December 2023 and January 2024 test sets combined, the dark vessel detector flagged 25 and 24 vessels respectively with anomaly score > 0.5 . All flagged vessels were fishing-registered, consistent with the known pattern that fishing vessels operating near regulatory boundaries are disproportionately likely to manipulate AIS [12]. In the absence of ground-truth dark-event labels (SAR/AIS fusion is required for true recall evaluation), precision cannot be formally computed; however, all flags correspond to plausible IUU fishing behaviour based on vessel type, speed, and geographic context.

H. Computational Performance

Table VI summarises wall-clock times for key operations.

GPU-accelerated XGBoost achieves a $\sim 22\times$ training speedup over sklearn RandomForest on equivalent hardware, enabling rapid model updates as new AIS data becomes available. Inference latency of 1.2 s for 64 k examples ($\approx 19 \mu\text{s}/\text{prediction}$) supports near-real-time operational deployment.

TABLE VI: Computational Performance (RTX 4080 + 32 CPU Cores)

Operation	Time	Backend
Feature extraction (216k examples)	47 s	32 CPU cores
XGBoost training (activity, 151k ex.)	38 s	GPU (CUDA)
XGBoost training (vessel type, 151k ex.)	44 s	GPU (CUDA)
Inference (64k test examples)	1.2 s	GPU (CUDA)
sklearn RF training (equivalent)	≈14 min	32 CPU cores

I. Regional GFW Validation

The synthetic simulation is validated against GFW 2023 real fleet-effort data for all four simulated regions. The Brazilian EEZ region illustrates the comparison: the simulator correctly reproduces the dominance of trawler effort in the southern high-seas zones and the seasonal peak in austral summer months (December–March). The Philippines EEZ simulation matches the GFW-observed dominance of squid-jigger and purse-seiner gear types. Quantitative comparison of simulated vs. real speed profiles using Kullback–Leibler divergence yields mean $D_{KL} < 0.15$ nats across all gear types and regions, indicating statistically plausible synthetic tracks.

VIII. DISCUSSION

A. EO/SAR Generalisation Gap

The most significant open challenge is performance on EO/SAR-only observations where nav_status is absent and SOG may be unavailable (EO) or imprecise (SAR Doppler). Our confidence calibration (Eq. 2) correctly assigns low confidence ($c \approx 0.30$) to single EO frames; however, the *predicted class* in those cases depends heavily on spatial context (proximity to fishing grounds, shipping lanes) and time of day. Recent work on SAR foundation models like SARatRX [?] demonstrates that pre-training on massive unlabeled target datasets can improve few-shot recognition of vessels, offering a path toward better EO/SAR-only classification. Future work should incorporate these foundation backbones alongside a maritime geography prior—encoding known shipping lanes, fishing zones, and anchorage areas as additional features—to improve single-observation accuracy for sensor-limited contacts.

B. Class Imbalance in Fishing Detection

Fishing vessels represent $< 1\%$ of AIS traffic in US coastal waters (the NOAA training domain), leading to imbalanced training data (240 fishing vs. 22,684 transit examples in the test set). Addressing this requires targeted augmentation from open-ocean datasets where fishing density is higher: the GFW fishing-hours data confirms that fishing effort in the Philippine EEZ is $8\times$ denser per km^2 than in US coastal waters. Incorporation of Pacific and Indian Ocean AIS data would substantially improve fishing detection recall.

C. MMSI Anonymisation in INFORE

The INFORE dataset uses UUID-based vessel identifiers rather than real MMSIs, preventing registry lookup or cross-referencing with GFW vessel data. As a result, vessel-type

labels for this corpus rely on the self-reported ITU field, which may be miscoded. This limits the utility of INFORE for vessel-type classifier evaluation.

D. Towards SAR-AIS Fusion

A production MDA system would fuse SAR vessel detections with AIS tracks in a track-association layer, providing both the kinematic signature (SAR gives position, heading, rough speed) and the identity claim (AIS provides MMSI). Our architecture is designed to accept both inputs simultaneously: a SAR ping and concurrent AIS pings for the same vessel can be merged into a single feature vector, with the sensor_enc feature distinguishing the origin of each measurement. Implementing this fusion layer is a priority for future work.

E. Limitations

- 1) **Geographic coverage.** All real training data are from US coastal waters and the Eastern Mediterranean. Performance in open-ocean and equatorial regions (Malacca, Guinea) is extrapolated via simulation; real validation requires satellite AIS data from those regions.
- 2) **Temporal coverage.** Three daily snapshots across six months provide limited seasonal diversity; monthly or quarterly bulk data would better capture seasonal fishing patterns.
- 3) **No STS or loiter ground truth.** Ship-to-ship transfer and loitering events are absent from the NOAA nav_status field and can only be evaluated via synthetic simulation. Real STS events require cross-referencing commercial AIS databases or incident reporting.
- 4) **Model interpretability.** SHAP analysis confirms that sog_mean, pct_slow, nav_status (when present), and loiter_index are the dominant activity discriminators, with the first two features together accounting for $> 60\%$ of mean absolute SHAP contribution. Full SHAP waterfall plots are available in the supplementary material.

IX. CONCLUSION

We have presented an end-to-end Maritime Activity Intelligence Engine that addresses the partial-track classification challenge central to multi-sensor maritime surveillance. By defining a 42-feature vector valid for any $N \geq 1$ observations, training XGBoost with native NaN handling on GPU hardware, and using ping-level nav_status as ground truth, the system achieves robust classification performance that remains statistically flat from single-observation EO contacts ($N = 1$) to full AIS tracks ($N = 50$).

Across 5,121 real vessels from three independent NOAA datasets, the activity classifier achieves macro $F_1 = 1.000$ where nav_status is available and weighted $F_1 = 0.900\text{--}0.995$ in cross-dataset evaluation. The vessel-type classifier achieves macro $F_1 = 0.969$ across 12 classes. GPU acceleration provides a $22\times$ training speedup over CPU-based RandomForest, enabling rapid retraining as new data becomes available.

The most significant future direction is integration of SAR/EO-derived vessel detections as first-class inputs: our

sensor-fusion architecture already accommodates these modalities, and satellite-based open-ocean coverage is the critical data gap limiting geographic generalisation.

APPENDIX A UNIFIED VESSEL TYPE TAXONOMY

Table VII lists the 12-class unified vessel taxonomy and the mapping from ITU AIS codes and GFW gear types.

TABLE VII: Unified Vessel Type Taxonomy

Unified Class	ITU Codes	GFW Gear Types
fishing	30	trawlers, drifting_longlines, set_longlines, squid_jigger, purse_seines, gillnets
cargo	70–79	—
tanker	80–89	—
passenger	60–69	—
tug	31–32	—
naval	35, 55	—
support_vessel	33–34, 50–54, 58–59	—
sailing	36	—
pleasure_craft	37	—
hsc	40–49	—
other	90–99	—
unknown	0	—

APPENDIX B SIMULATION PARAMETERS

Each simulated vessel template specifies: speed range (kn) per activity phase, dark probability per hour, heading variance, AIS reporting interval, and physical dimensions (length, beam, draught). Fishing vessels use a five-phase voyage model: {transit-to-ground, fishing, transit-between-sets, loitering, return-to-port}. Cargo and tanker vessels use a three-phase model: {port-departure, oceanic-transit, port-approach}. Parameters are fitted from GFW vessel-registry length/speed distributions and MMSI-activity data.

ACKNOWLEDGMENTS

The authors thank Global Fishing Watch for making their vessel-effort and vessel-registry data publicly available under CC-BY-NC 4.0 [3], NOAA Office for Coastal Management for the MarineCadastre AIS dataset [4], and the INFORE project for the Piraeus AIS dataset [5].

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